Karla L. Sahin

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EDUCATION

Carnegie Mellon University - Pittsburgh, PA *May 2028*
B.S. Electrical and Computer Engineering; Intended Additional Major in Robotics **GPA: 3.85, Dean's List**

Relevant Coursework: Embedded Systems, Computer Vision, Multi-Robot Planning and Coordination (PhD-level), Electronic Devices and Analog Circuits (TA), Planning Techniques for Robotics, Structure and Design of Digital Systems

SKILLS

Languages: C++, Python, C, SystemVerilog, MATLAB, JavaScript, Bash, LaTeX, ARM Assembly
Robotics & Software: ROS 2, RTOS, Gazebo, OpenCV, Docker, Linux, Git, CI/CD, GDB, Grafana, pytest, A*, RRT/PRM, MAPF
Hardware: FPGA (Vivado), PCB Design, Arduino, Raspberry Pi, STM32 (ARM Cortex-M4), NVIDIA Orin, UART/I2C/ADC, SOLIDWORKS, Fusion 360, Oscilloscope/DMM/Soldering

EXPERIENCE

Robotics Software Test Engineering Intern - Locus Robotics - Nashua, NH May 2026 - Present

- Built ROS 2 hand-eye calibration validation for a 6-DoF picking arm and multi-camera rig, quantifying camera-to-kinematics agreement via PnP, fiducial detection, and TF composition; utilized Hardware-in-the-Loop (HIL) test benches with NVIDIA Orins
- Developed reprojection error and point cloud overlay visualization tools in OpenCV for diagnosing calibration drift without recalibration or added hardware
- Shipped Grafana/VictoriaMetrics dashboards tracking network health and Locus Array's pick performance fleet-wide at customer sites; integrated controller board telemetry into ROS for lighting control
- Automated state machine diagram generation in CI to keep architecture docs in sync with code, and integrated arm and camera rig into Gazebo sim for hardware-free development

Teaching Assistant (Lab Lead) - Carnegie Mellon - 18-220 Electronic Devices and Analog Circuits January 2026 - Present

- Lead 2 weekly recitations, facilitate 3-hour labs, and host office hours to guide students through course content

HCI Researcher - MIT (CSAIL) Human-Computer Interaction Engineering Lab - Cambridge, MA May - August 2025

- Virtually modeled tactile and sensing feedback of assembly-free multi-material 3D-printed structures (Xspine) for original or Minecraft force-tunable game controllers and fidget toys with Python, Git, Unity, and React
- Architected an interface to process 8 high-frequency joint signals (100 Hz), generating data for AI models

PROJECTS - *Portfolio: https://klsahin.github.io/*

Space Planning for Assembly and Repair - *Multi-Robot Planning and Coordination* Present

- Designing a decentralized multi-robot planner for in-space assembly, combining distributed priority negotiation, kinodynamic RRT* path planning, and reactive fallback control to handle unreliable localization and intermittent communication
- Building an execution-time repair system with RL-guided task reassignment, retiming, and rerouting for a free-flying robot fleet operating under communication loss, sensor drift, and partially known environments

Bare-Metal ARM Kernel - *Embedded Systems* Present

- Built a lightweight preemptive RTOS kernel from scratch for STM32 (ARM Cortex-M4), implementing context switching, priority-based scheduling, and task synchronization primitives (mutexes/semaphores)
- Wrote assembly bootloader, user/kernel privilege separation via SVC syscall handlers, interrupt-driven UART ring buffers, and hardware drivers (I2C, ADC, LCD)

High-DoF Robotic Arm Motion Planning - *Planning Techniques for Robotics* May 2026

- Implemented RRT, RRT-Connect, RRT*, and PRM motion planners in C++ for robotic arms with up to 7 DoF
- Built a collision-avoidance system coordinating two UR5e arms to pick, hand off, and place an object, simulated in PyBullet

Hardware-Accelerated RTL System - *Digital Systems* April 2026

- Designed an RTL system in SystemVerilog and Vivado for high-speed game logic and real-time feedback on FPGA
- Built an FSM-based control path and custom datapath for 12-bit input processing, nanosecond determinism at 40-200 MHz

PUBLICATIONS

- Mingming Li, **Karla Sahin***, Haotian Chen*, Dingning Cao*, Stefanie Mueller, and Jiaji Li (*equal contribution). "Xspine: Integrating Motion Sensing Capability into Dynamic Structures Using Multi-material FDM 3D Printing." ACM CHI 2026 (Full Paper & Interactive Demo); ACM UIST 2025 (Poster). DOI: 10.1145/3772318.3791317

ACTIVITIES & LEADERSHIP

NASA Lunabotics Competition Team (CMU) - *Software & Perception Team* September 2026 – Present
ChemE Cube Team (CMU) - *Senior Build Lead* January 2025 – Present

- Led weekly build sessions for Direct Air Capture system, winning 5th-place out of 54 teams at 2025 AIChE Annual Conference

Society of Women Engineers (CMU) - *SWE++ Curriculum Lead* January 2025 - Present

- Directed 8-week Python camp: developing lessons/starter code, managing volunteers, and teaching 20+ middle schoolers